



University of
Nottingham
UK | CHINA | MALAYSIA

COMP3055

Machine Learning

Topic 2 – Design a Learning System

2025 Autumn

Human Learning vs Machine Learning

- Cognitive science vs computational science
 - Animal learning vs machine learning
 - Don't fly like birds
 - Many ML models are based on human types of learning
- Evolution vs machine learning
 - Adaptation vs learning

对比项	人类学习	机器学习
学科领域	认知科学 (Cognitive Science)	计算科学 (Computational Science)
模式	动物/人类的神经学习机制	数学建模与算法优化
模拟方式	“不要像鸟一样飞”——不是模仿，而是借鉴	以人类学习为灵感，抽象为计算模型
进化 vs 学习	进化是种群层面长期适应	机器学习是个体系统通过经验改进性能

Motivating Problems

Face Recognition



Motivating Problems

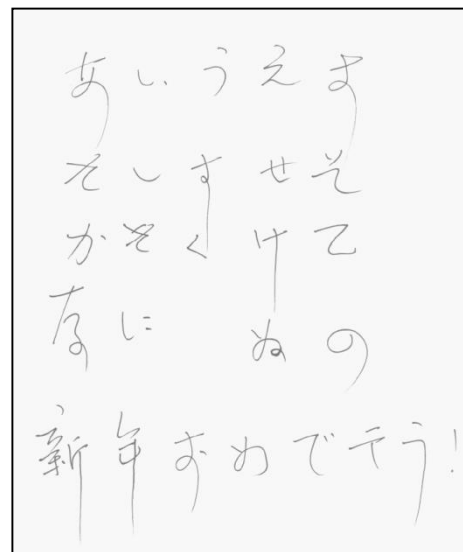
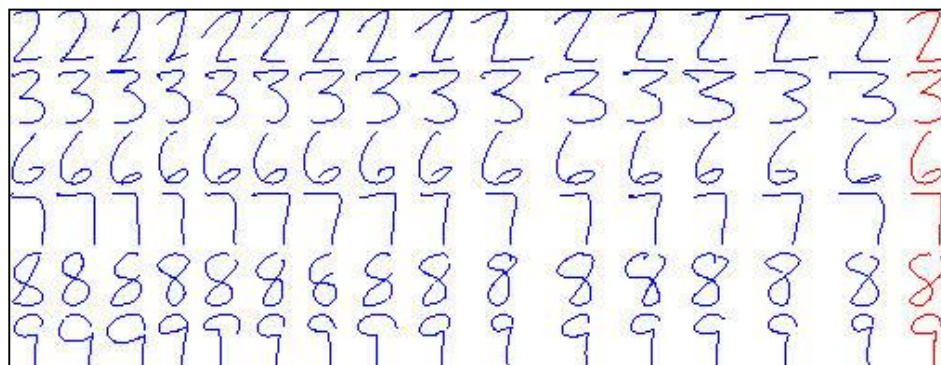
指纹识别：如边境控制

Fingerprint Recognition (e.g., border control)



Motivating Problems

Handwritten Character Recognition



完全非限定脱机手写汉字的识别
完全非限定脱机手写汉字的识别
完全非限定脱机手写汉字的识别

Motivating Problems

DaVinci外科手术机器人

DaVinci surgical robot by Intuitive Surgical.

St. Elizabeth Hospital is one of the local hospitals using this robot. You can see this robot in person during an open house ([website](#)).



日本HAL外骨骼健康辅助系统

Japanese health care assistant suit
(HAL - Hybrid Assistive Limb)



Also... Mind-
controlled wheelchair
using NI LabVIEW

思维控制轮椅

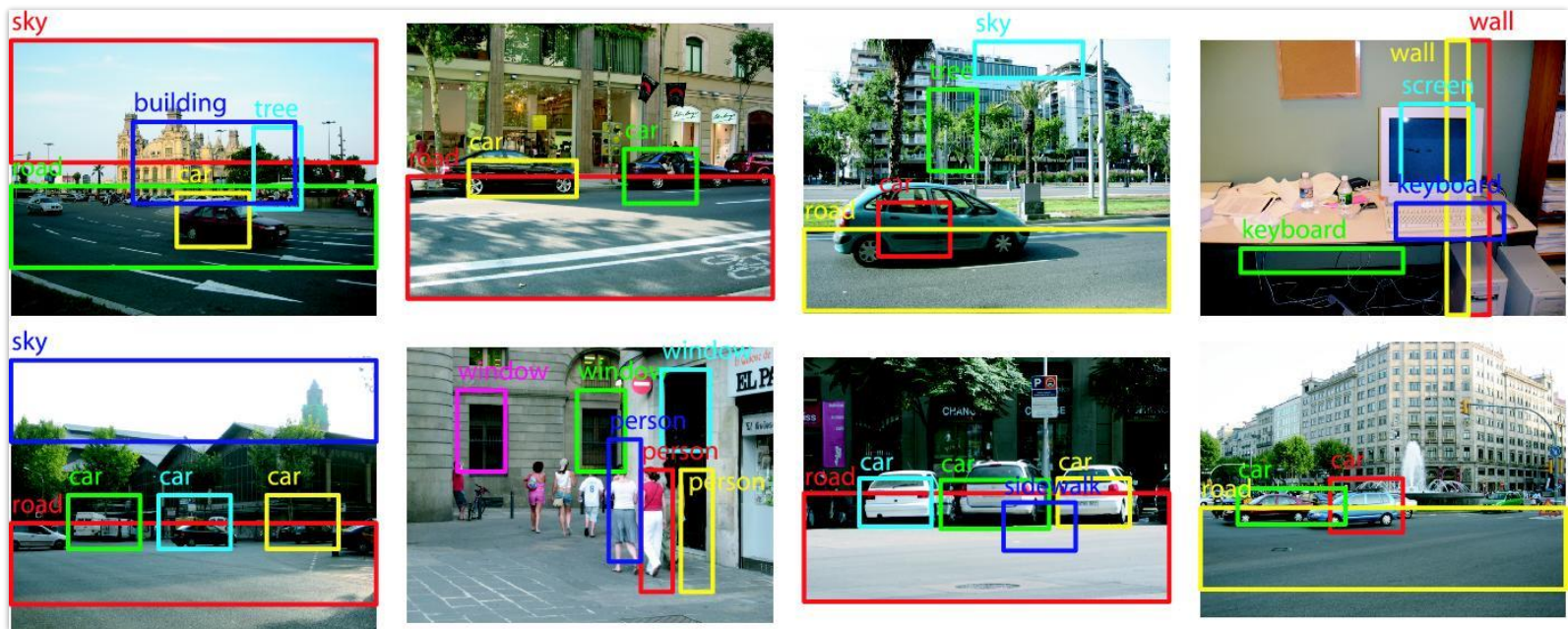
Motivating Problems

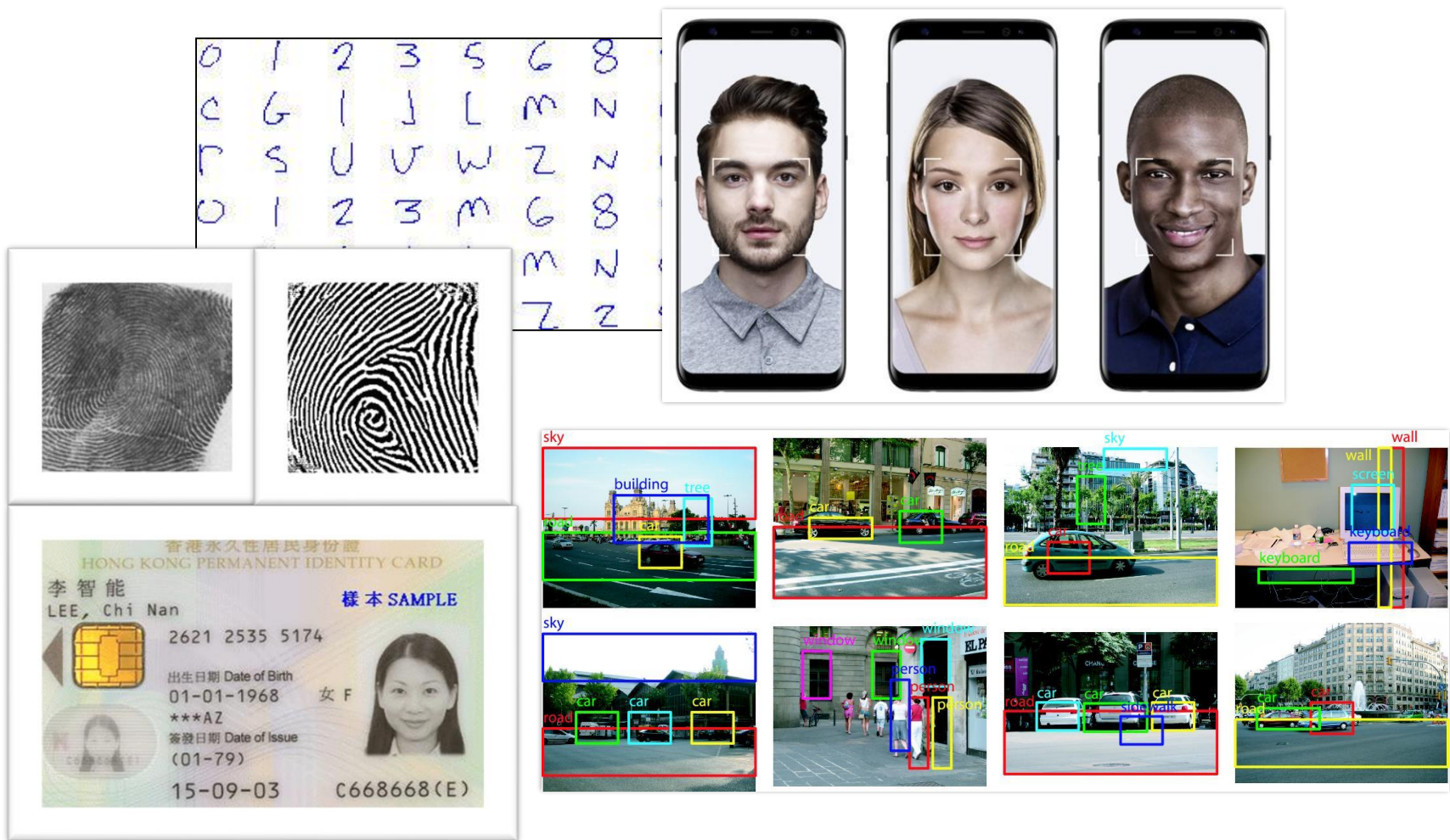
- **Space Robot** – Mars Autonomous navigation features with human remote control and oversight



Motivating Problems

Object Recognition





...

Can Machines Learn to Solve These Problems?

Definition of Learning

A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .

-- from Mitchell, Machine Learning, McGraw-Hill, 1997

如果一个计算机程序在针对某一类任务 T 的执行中，通过经验 E 所取得的性能指标 P 的表现能够随着经验 E 的积累而有所提升，那么就可以说该程序是从经验 E 中学习的。

Definition of Learning

What does this mean exactly?

For example, handwriting recognition problem

- Task ***T***: Recognizing hand written characters
- Performance measure ***P***: percent of characters correctly classified
- Training experience ***E***: a database of handwritten characters with given classifications

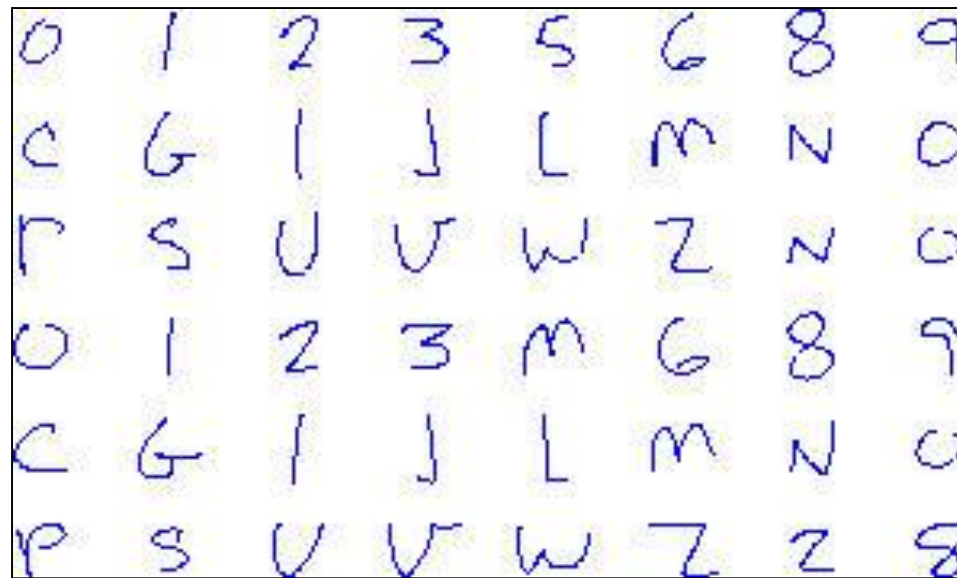
要素	含义
T (Task)	识别手写字符
P (Performance Measure)	正确分类率 (accuracy)
E (Experience)	已标注的手写数字数据库 (如 MNIST)



Definition of Learning

What are design issues and approaches?

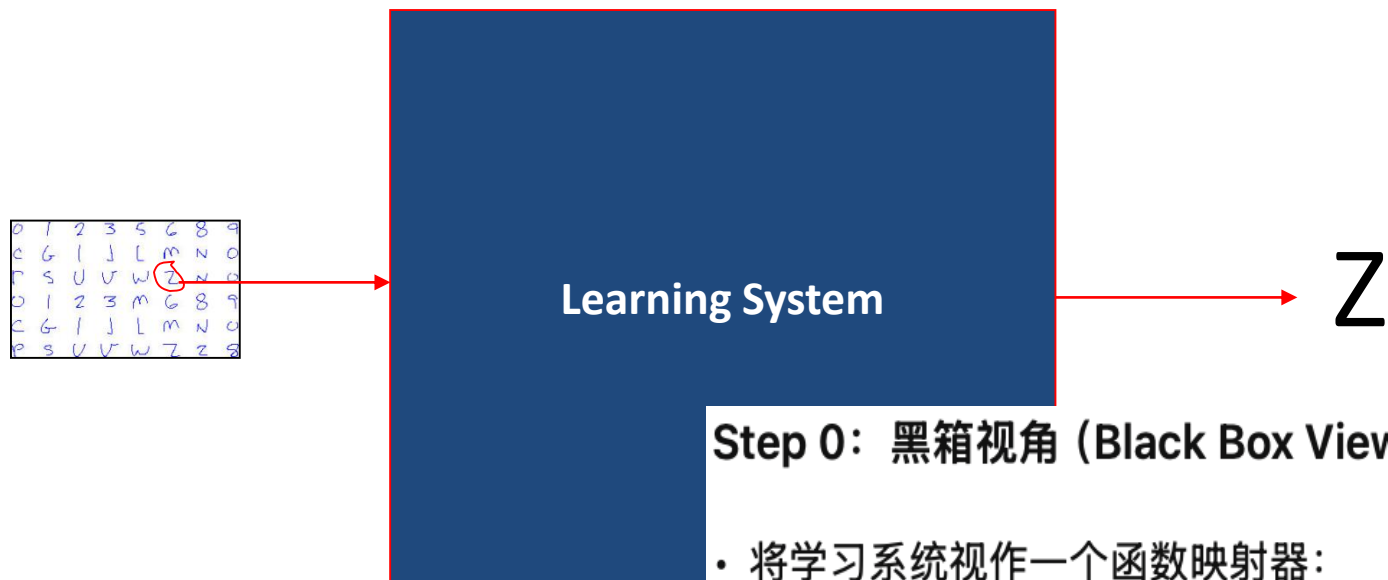
For example, handwriting recognition problem



Design a Learning System

Step 0:

- Lets treat the learning system as a black box



Step 0: 黑箱视角 (Black Box View)

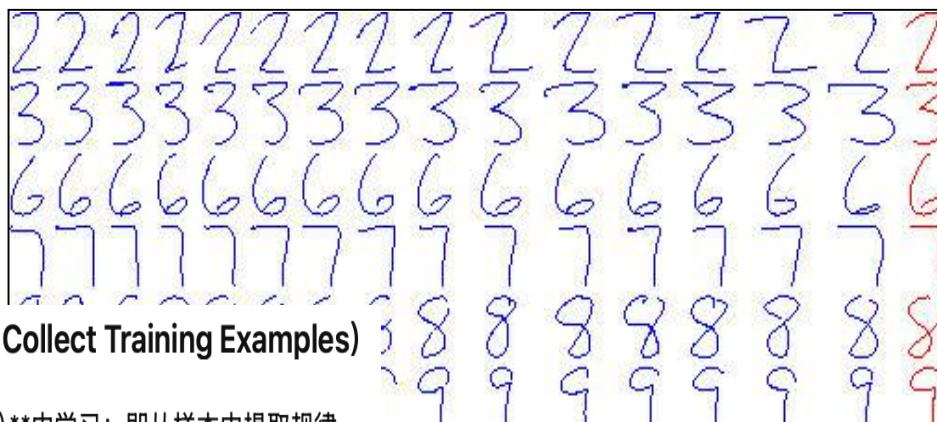
- 将学习系统视作一个函数映射器：
输入 → 系统 → 输出
- 我们要设计的系统 Z 能“通过数据学习映射关系”。

Design a Learning System

Step 1: Collect Training Examples (Experience).

- Without examples, our system will not learn
 - so-called learning from examples

Identify or Class or Label



2
3
6
7
8
9

Step 1: 收集训练样本 (Collect Training Examples)

- 系统从**经验 (Experience)**中学习：即从样本中提取规律。
- 这是“learning from examples”的核心。
- 没有数据，系统无法学习。

Design a Learning System

Step 2: Representing Experience

- Choose a representation scheme for the experience / examples

Step 2: 表示经验 (Representing Experience)

1 特征表示 (Feature Representation)

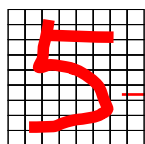
- 将传感器输入转换为特征向量:

$$X = (x_1, x_2, x_3, \dots, x_n)$$

- 每个样本都是 n 维向量, 反映输入的特征值。

示例:

手写数字识别 $\rightarrow$ 28x28 灰度图 $\rightarrow$ 784 维向量。



$$X = (1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 1, 1, \dots, 1); 64\text{-d v}$$

The sensor input represented by an n -d vector,
called the **feature vector**, $X = (x_1, x_2, x_3, \dots, x_n)$

Design a Learning System

Step 2: Representing Experience

- THE MNIST DATABASE

<http://yann.lecun.com/exdb/mnist/>

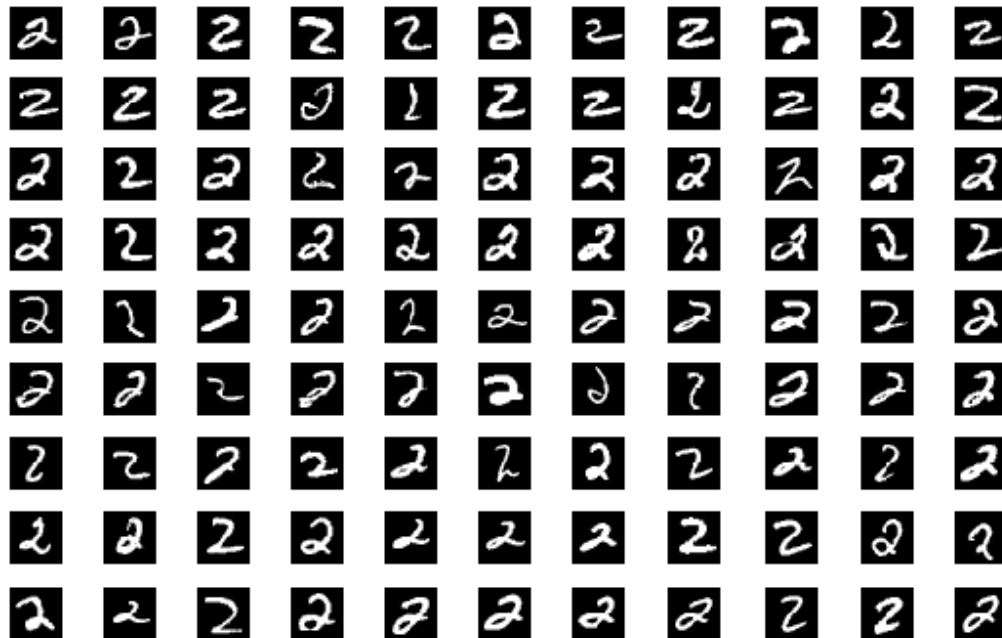
- The original black and white (bi-level) images from NIST were size normalized to fit in a 20x20 pixel box while preserving their aspect ratio. The resulting images contain grey levels as a result of the anti-aliasing technique used by the normalization algorithm. The images were centered in a 28x28 image by computing the center of mass of the pixels, and translating the image so as to position this point at the center of the 28x28 field.

Design a Learning System

Step 2: Representing Experience

- THE MNIST DATABASE

<http://yann.lecun.com/exdb/mnist/>



The feature vector of input data is a 784 dimensional vector

Design a Learning System

为以下“体验/示例”选择一种表示方案：
由一个 n 维向量表示的传感器输入，该向量被称为特征向量，记为 $X = (x_1, x_2, x_3, \dots, x_n)$ 。

Step 2: Representing Experience

- Choose a representation scheme for the experience/examples

要表示这种体验，我们需要知道 X 是什么。

因此，我们需要一个相应的向量 D ，它将记录我们关于 X 的知识（体验）。

- The sensor input represented by an n -d vector, called the feature vector, $X = (x_1, x_2, x_3, \dots, x_n)$
体验 E 是一个由两个向量组成的对，即 $E = (X, D)$
- To represent the experience, we need to know what X is.
- So we need a corresponding vector D , which will record our knowledge (experience) about X .
- The experience **E is a pair of vectors $E = (X, D)$.**

Design a Learning System

Step 2: Representing Experience

- Choose a representation scheme for the experience/examples.
 - The experience $\mathbf{E}$ is a pair of vectors $\mathbf{E} = (\mathbf{X}, \mathbf{D})$.
- So, what would $\mathbf{D}$ be like? There are many possibilities.

1. $\mathbf{E} = (\mathbf{X}, \mathbf{D})$ 的含义

- **E (Experience)**: 一次“学习经验”或“训练样本”

它是由两个部分组成的:

$$\mathbf{E} = (\mathbf{X}, \mathbf{D})$$

- **X**: 输入特征 (feature vector)

表示系统“看到”的内容，比如一张图片的像素值、一个声音信号的频谱、或一个文本的向量编码。

- **D**: 输出目标 (desired output / label)

表示系统“应该输出什么”或“正确答案是什么”，由人类或外部系统提供。

 换句话说:

模型通过学习大量 $(\mathbf{X}, \mathbf{D})$ 的配对样本，逐渐学会从 $\mathbf{X}$ 预测出正确的 $\mathbf{D}$ 。

Design a Learning System

Step 2: Representing Experience

- So, what would **D** be like? There are many possibilities.
- Assuming our system is to recognise 10 digits only, then **D** can be a 10-d binary vector; each correspond to one of the digits.

$D = (d_0, d_1, d_2, d_3, d_4, d_5, d_6, d_7, d_8, d_9)$

e.g.,

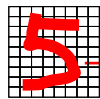
if X is digit 5, then $d_5=1$; all others $=0$

Design a Learning System

Step 2: Representing Experience

- So, what would **D** be like? There are many possibilities.
- Assuming our system is to recognise 10 digits only, then **D** can be a 10-d binary vector; each correspond to one of the digits.

$D = (d_0, d_1, d_2, d_3, d_4, d_5, d_6, d_7, d_8, d_9)$



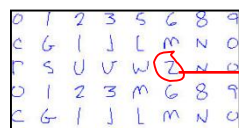
$X = (1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 1, 1, \dots, 1)$; 64-d Vector

$D = (0, 0, 0, 0, 0, 1, 0, 0, 0, 0)$

Design a Learning System

Step 3: Choose a Representation for the Black Box

- We need to choose a function F to approximate the block box. For a given X , the value of F will give the classification of X . There are considerable flexibilities in choosing F .



X

Learning System

F

$F(X)$

在机器学习中，我们把“学习系统”视为一个黑箱 (black box)。我们不知道它内部是怎么工作的，但我们知道它的输入 X 和输出 $F(X)$ 。

我们的任务是：选择一个合适的函数 F 来表示这个黑箱，让它能根据输入 X 产生正确的输出 (也就是我们希望的 D)。

换句话说：

机器学习的本质就是找到一个合适的函数 F ，使得 $F(X) \approx D$ 。

Design a Learning System

Step 3: Choose a Representation for the Black Box

- F will be a function of some adjustable parameters, or weights, $W = (w_1, w_2, w_3, \dots, w_N)$, which the learning algorithm can modify or learn

一、核心思想

学习系统 F 不仅仅是一个固定的函数，而是一个由可调整参数 (weights) 控制的函数。

这些参数统称为 W (权重向量, weights)，学习的过程其实就是不断修改 W ，使得系统 $F(W, X)$ 对输入 X 的预测结果越来越接近正确答案 D 。



X

Learning System

$F(W, X)$

$F(W)$

符号

含义

F

学习系统 (模型)

X

输入 (feature vector)

W

参数 (权重向量)

$F(W, X)$

模型输出，取决于输入 X 与当前参数 W

Design a Learning System

Step 4: Learning/Adjusting the Weights

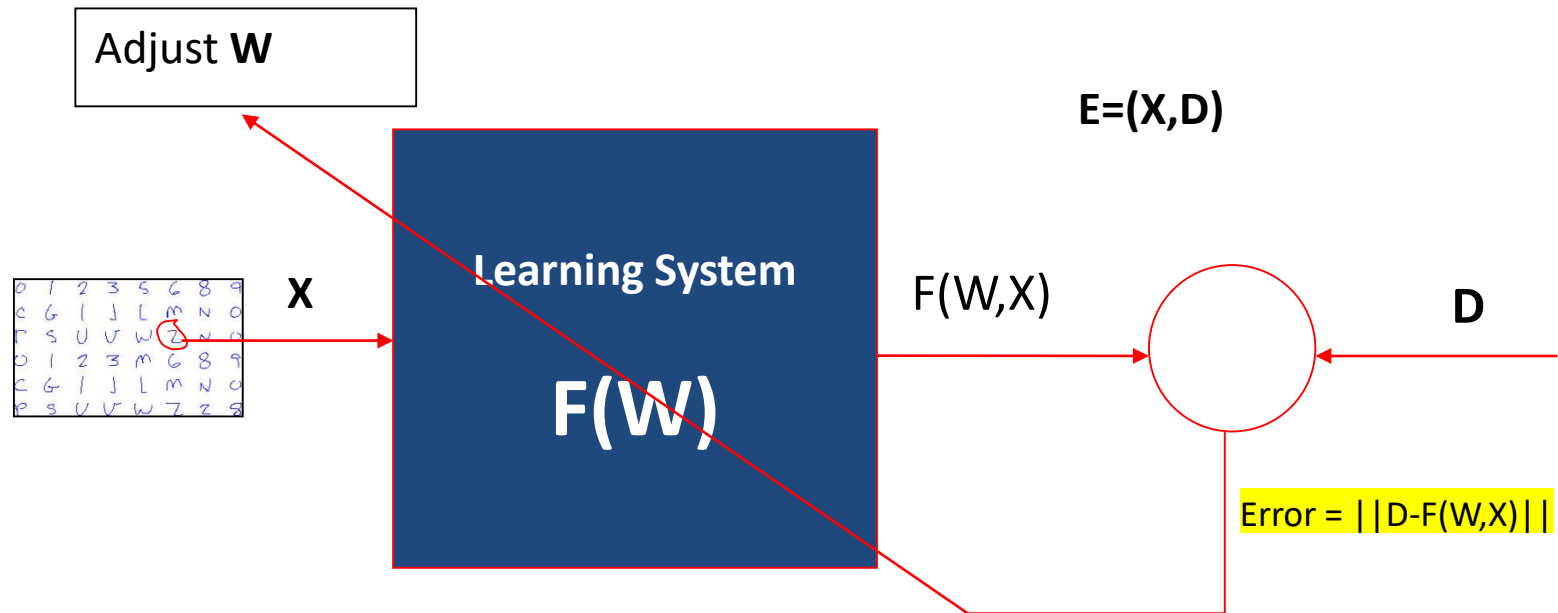
- We need a learning algorithm to adjust the weights such that the experience/prior knowledge from the training data can be learned into the system:

$$E=(X,D)$$

$$F(W,X) = D$$

Design a Learning System

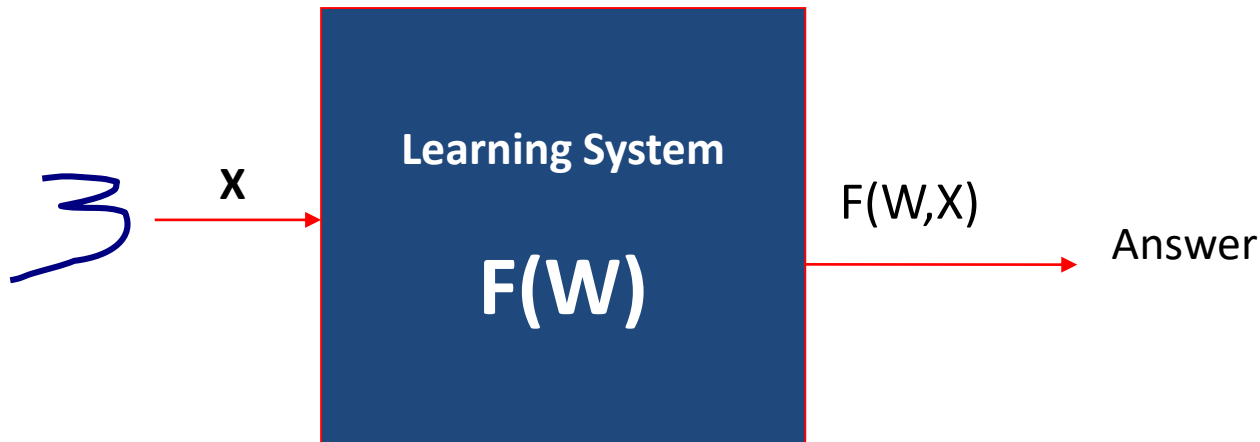
Step 4: Learning/Adjusting the Weights



Design a Learning System

Step 5: Use/Test the System

- Once learning is completed, all ^{参数}parameters are fixed. An unknown input $\mathbf{X}$ is presented to the system, the system computes its answer according to $\mathbf{F}(\mathbf{W}, \mathbf{X})$

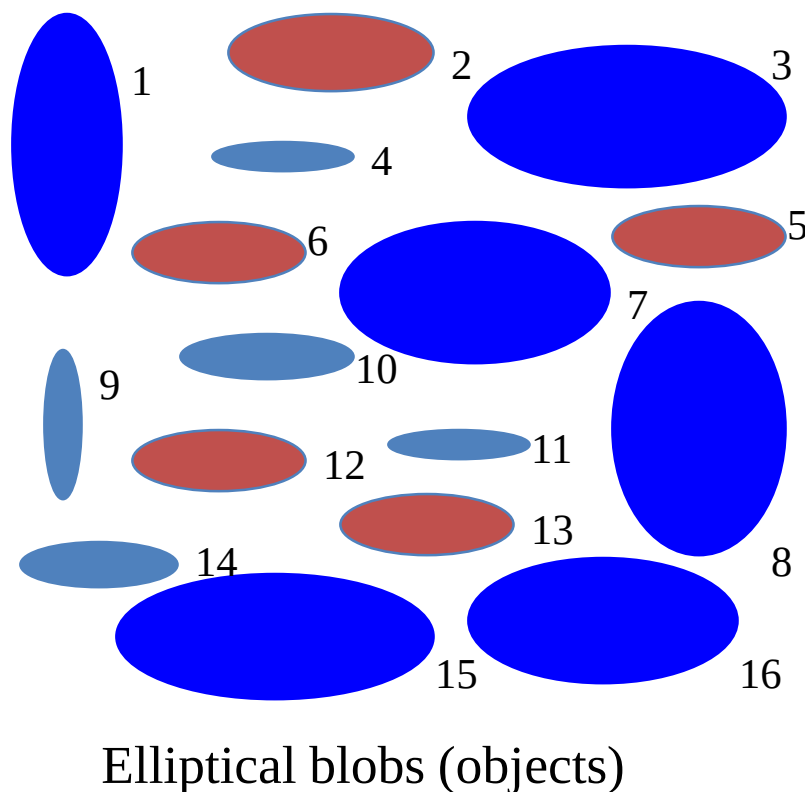
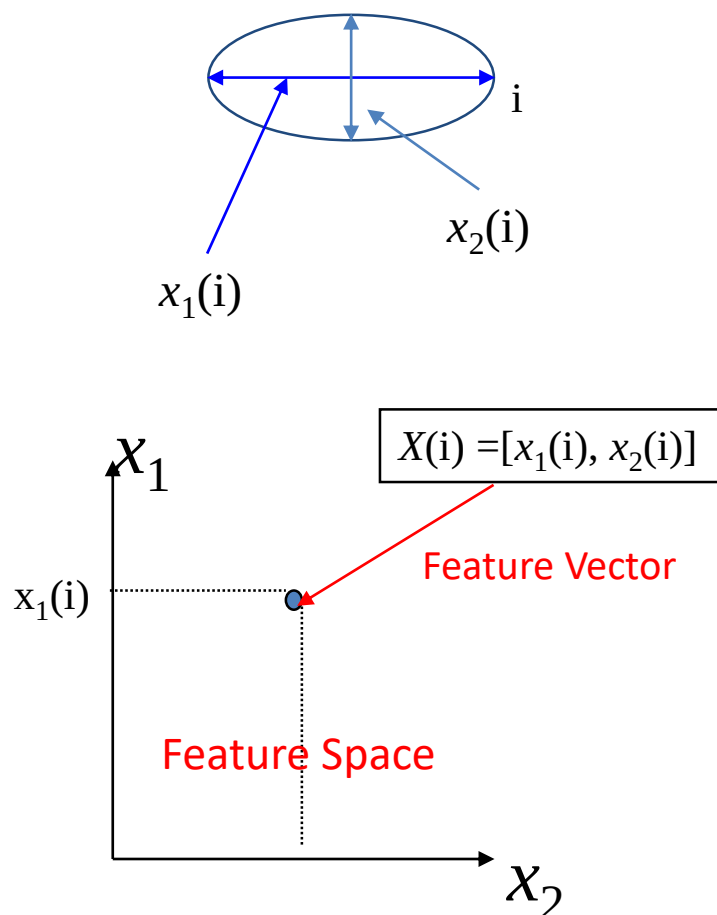


1 定义

- 特征向量 X 代表一个样本点。
- 所有样本的特征构成一个 n 维空间。
- 在特征空间中，每个样本是一个点。

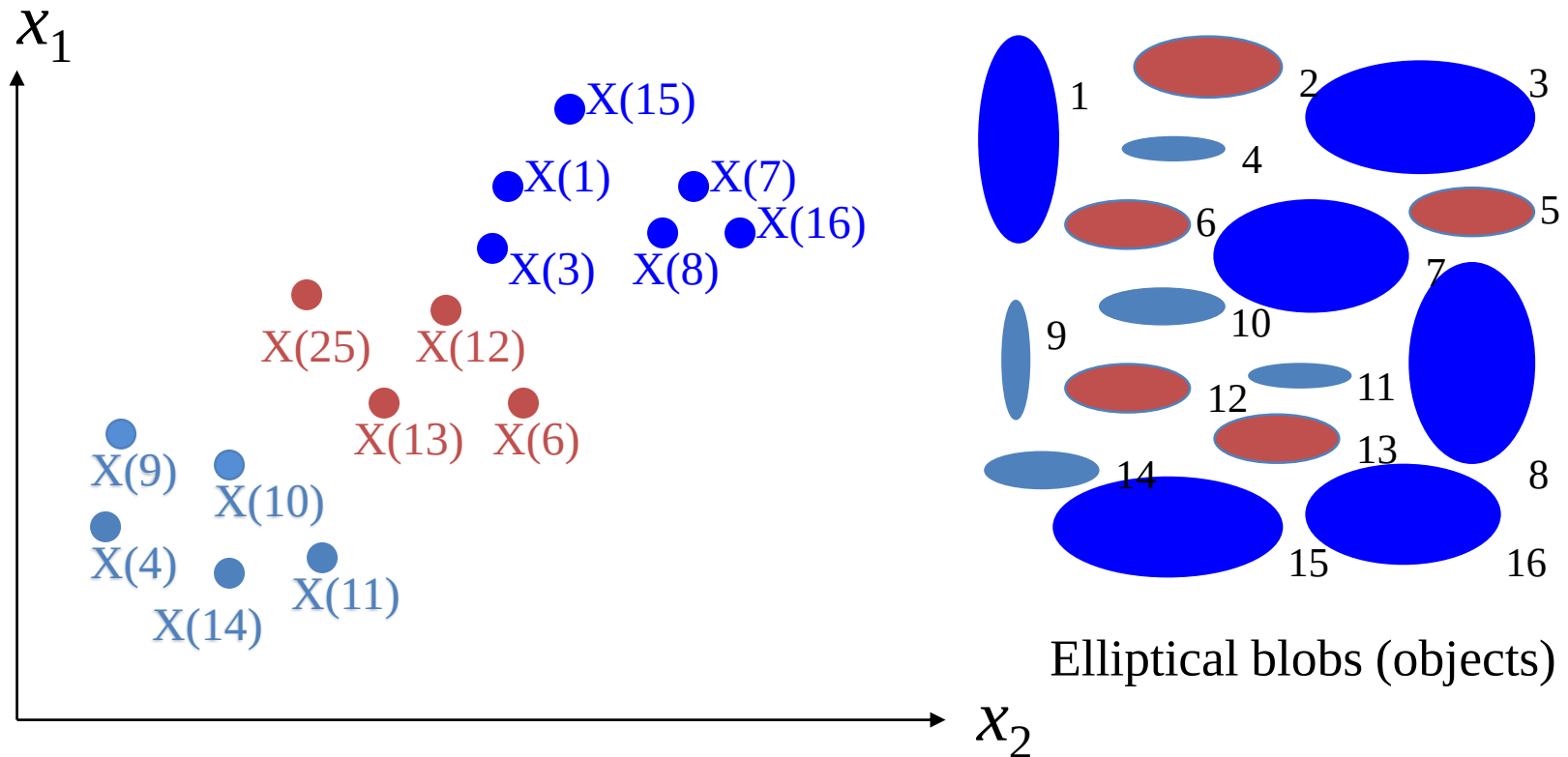
Feature Space

Representing real world objects using feature vectors



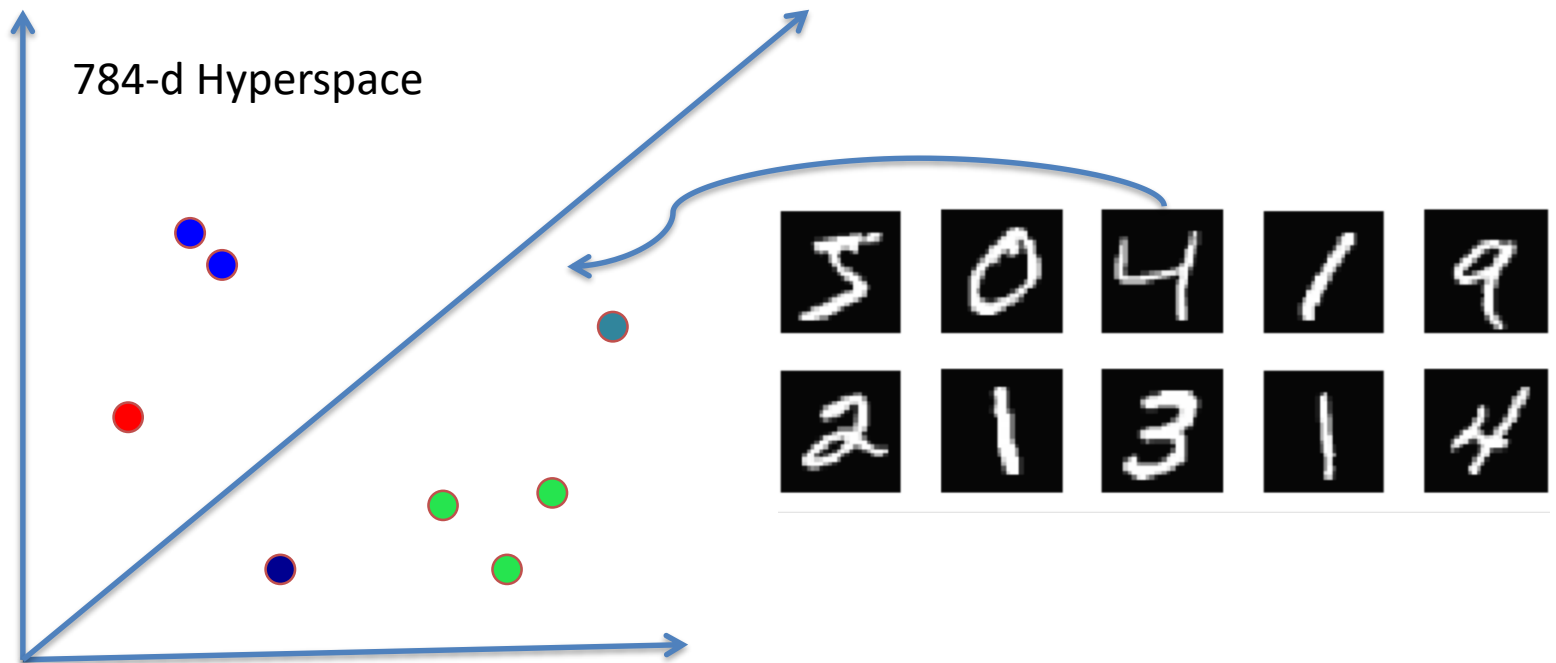
Feature Space

From **Objects** to **Feature Vectors** to **Points** in the **Feature Spaces**



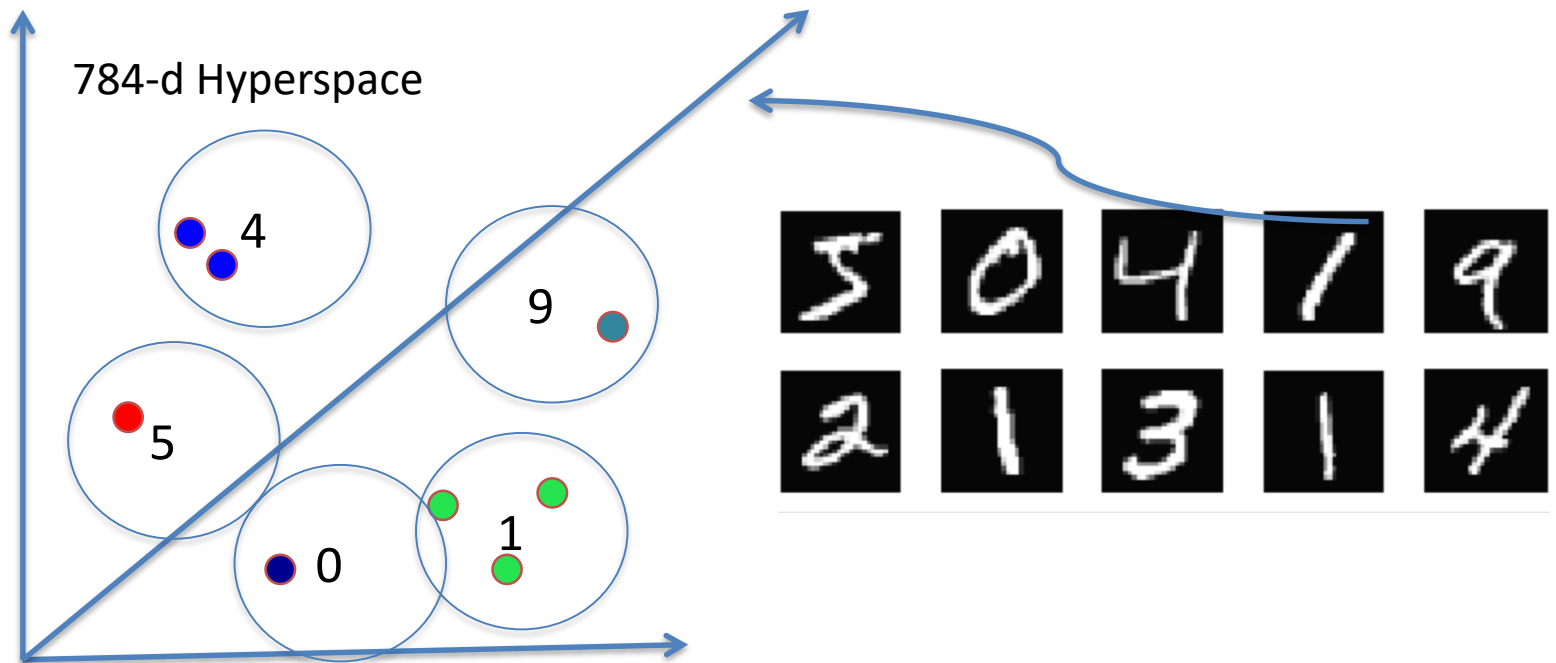
Feature Space

From **Objects** to **Feature Vectors** to **Points** in the **Feature Spaces**



Feature Space

From **Objects** to **Feature Vectors** to **Points** in the **Feature Spaces**



Representing General Objects

Feature vectors of

Faces

Cars

Fingerprints

Gestures

Emotions (a smiling face, a sad expression etc)

...

对象类型

特征表示示例

人脸

面部几何比例、像素特征

汽车

形状、颜色、轮廓特征

指纹

纹线方向、特征点

手势

手部关键点坐标

情绪

面部表情参数、语音特征

Further Reading

Chapter 1, T. M. Mitchell, Machine Learning, McGraw-Hill International Edition, 1997

Any Questions?

